

Binary-Outcome LPS Progress Report

Source path: ~/current_projects/geosmooth/split_handoffs/
lps_binary_outcome_progress_2026-06-07/lps_binary_outcome_progress_report.tex

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Purpose

This report records the current binary-response extensions of the local polynomial smoother (LPS) implemented in `geosmooth`. The implementation now has two binary modes:

```
outcome.family = "bernoulli"    and    outcome.family = "binomial".
```

Both modes estimate a probability surface, but they use different local fitting objectives. The first is a minimal conditional-expectation/Brier-risk mode built on the ordinary least-squares LPS core. The second is a principled local weighted logistic-polynomial mode.

The report has three goals:

1. define both implemented binary modes precisely;
2. describe the diagnostics already exposed by the package;
3. freeze an initial experimental design for comparing the two modes.

The results of the proposed experiments will be added in a later revision.

1 Binary Target

For binary outcomes,

$$Y_i \in \{0, 1\},$$

the conditional expectation is also the event probability:

$$p(x) = \mathbb{E}(Y \mid X = x) = \mathbb{P}(Y = 1 \mid X = x).$$

Therefore, a binary-response LPS fit should be interpreted as an estimate

$$\hat{p}(x) \approx p(x).$$

This is different from a continuous-response smoother only in interpretation and loss function. For continuous responses, the fitted value estimates an unconstrained regression surface. For binary responses, the fitted value should live on the probability scale:

$$0 \leq \hat{p}(x) \leq 1.$$

2 Ordinary LPS Setup

Let $x_1, \dots, x_n \in \mathbb{R}^D$ be observed covariates and let y_i be the response. At a prediction point x , LPS selects a local support

$$N(x; k) \subset \{1, \dots, n\}, \quad |N(x; k)| = k,$$

where k is a candidate support size. Kernel weights

$$w_j(x; k, K) \geq 0, \quad j \in N(x; k),$$

are computed from distances between x and the support points using kernel K . A local polynomial feature vector is denoted by

$$\phi_x(x_j) \in \mathbb{R}^q.$$

For example, degree 0 uses only an intercept, degree 1 uses an intercept and local linear terms, and degree 2 adds local quadratic and interaction terms. The current binary work uses this same local-polynomial infrastructure.

3 `outcome.family = "bernoulli"`: Brier/Conditional-Expectation Mode

The `"bernoulli"` mode keeps the ordinary weighted least-squares local fit. At prediction point x , it solves

$$\hat{\beta}_x = \arg \min_{\beta} \frac{1}{2} \sum_{j \in N(x; k)} w_j(x; k, K) \{y_j - \phi_x(x_j)^\top \beta\}^2.$$

The raw prediction is

$$\hat{p}_{\text{raw}}(x) = \phi_x(x)^\top \hat{\beta}_x.$$

Because least-squares local polynomials are not probability models, the raw value may fall outside $[0, 1]$. The reported probability is therefore clipped:

$$\hat{p}(x) = \min\{1, \max\{0, \hat{p}_{\text{raw}}(x)\}\}.$$

The natural loss for this mode is the Brier score. For one observation,

$$\text{Brier}(y_i, \hat{p}_i) = (y_i - \hat{p}_i)^2.$$

For a held-out fold H , the cross-validated Brier score is

$$\text{CV}_{\text{Brier}} = \frac{1}{|H|} \sum_{i \in H} (y_i - \hat{p}_i)^2.$$

Interpretation

This mode is intentionally minimal. It estimates the conditional expectation $p(x)$ using the same local least-squares machinery used for numeric responses. It is fast, simple, and directly comparable with ordinary LPS. Its main weakness is that probability validity is enforced after fitting by clipping rather than by the local objective itself.

4 `outcome.family = "binomial": Local Logistic-Polynomial Mode`

The `"binomial"` mode fits a local Bernoulli model. Instead of modeling the probability directly by an unconstrained polynomial, it models the log odds:

$$\eta_x(x_j) = \text{logit}\{p_x(x_j)\} = \phi_x(x_j)^\top \theta_x,$$

where

$$\text{logit}(p) = \log \frac{p}{1-p} \quad \text{and} \quad \text{expit}(\eta) = \frac{1}{1 + \exp(-\eta)}.$$

Thus

$$p_x(x_j) = \text{expit}\{\phi_x(x_j)^\top \theta_x\}.$$

At prediction point x , the local logistic fit solves the weighted Bernoulli negative log-likelihood problem

$$\hat{\theta}_x = \arg \min_{\theta} \left[- \sum_{j \in N(x;k)} w_j(x; k, K) \{y_j \log p_{xj}(\theta) + (1 - y_j) \log(1 - p_{xj}(\theta))\} \right],$$

with

$$p_{xj}(\theta) = \text{expit}\{\phi_x(x_j)^\top \theta\}.$$

The fitted probability at the prediction point is

$$\hat{p}(x) = \text{expit}\{\phi_x(x)^\top \hat{\theta}_x\}.$$

Candidate selection uses observed cross-validated log loss. For numerical stability, fitted probabilities are truncated away from exact 0 and 1 inside the log-loss calculation:

$$\tilde{p}_i = \min\{1 - \varepsilon, \max\{\varepsilon, \hat{p}_i\}\}.$$

The held-out log loss is

$$\text{CV}_{\log\text{loss}} = -\frac{1}{|H|} \sum_{i \in H} \{y_i \log \tilde{p}_i + (1 - y_i) \log(1 - \tilde{p}_i)\}.$$

Interpretation

This mode is statistically more principled for binary data. The fitted values are probabilities by construction, and candidate selection uses a Bernoulli likelihood score. Its main risks are computational and numerical: local logistic fits can be unstable when a support has too few events, too few non-events, high-degree designs, or near-separation.

5 Implementation Status

Both modes are implemented in

`~/current_projects/geosmooth/R/lps.R.`

The user-facing entry point is

`fit.lps()`.

The relevant argument is

```
outcome.family = c("gaussian", "bernoulli", "binomial").
```

The current implementation enforces the following contract:

- **"gaussian"** preserves ordinary numeric-response LPS.
- **"bernoulli"** validates that responses are binary 0/1, uses the ordinary least-squares LPS core, reports clipped probability-scale fitted values, and stores unclipped raw fitted values.
- **"binomial"** validates that responses are binary 0/1, uses local weighted logistic-polynomial fits, routes `backend = "auto"` to the R backend, and rejects native backends for now.
- **"binomial"** candidate selection is based on observed CV log loss rather than observed CV RMSE.
- A candidate grid with no finite selection score fails fast rather than silently selecting an invalid candidate.

6 Diagnostics

For binary modes, the fitted object records probability diagnostics. Let $\hat{p}_i$ denote the reported probability and $\hat{p}_{\text{raw},i}$ the raw fitted value. In **"bernoulli"** mode, $\hat{p}_{\text{raw},i}$ is the unclipped least-squares prediction. In **"binomial"** mode, raw and reported fitted values are both on the probability scale, up to numerical clipping in loss calculations.

The probability diagnostics include:

- raw fitted-value range,

$$\min_i \hat{p}_{\text{raw},i}, \quad \max_i \hat{p}_{\text{raw},i};$$

- clipped probability range,

$$\min_i \hat{p}_i, \quad \max_i \hat{p}_i;$$

- fraction below 0 and above 1 before clipping;
- observed Brier score;
- observed log loss.

For **"binomial"** mode, the fit also records local logistic telemetry. The telemetry is split into CV fitting and final selected fitting. For each stage it reports:

- number of attempted local logistic solves;
- number and fraction converged;
- fallback-path count;
- event-rate fallback count;
- NA failure count;
- status counts.

This split is important because a broad experiment should not interpret a nominally logistic fit without knowing whether local predictions came from converged logistic solves, event-rate fallbacks, or failures.

7 Current Smoke-Test Status

A focused smoke script has already been run:

```
~/current_projects/geosmooth/scripts/run_lps_binary_outcome_smoke.R.
```

The generated HTML report is:

```
~/current_projects/geosmooth/split_handoffs/  
lps_binary_outcome_smoke_2026-06-07/lps_binary_outcome_smoke_report.html.
```

The latest audited smoke result accepted the implementation for broader binary experiment grids. The smoke telemetry showed that CV fitting used a mixture of converged logistic solves and fallback paths, while the final selected fit in that smoke example used converged logistic solves for all final local fits.

This smoke result is not yet a performance comparison. It is an implementation sanity check showing that both binary modes run, report probability diagnostics, and expose the logistic failure/fallback accounting needed for broader tests.

8 Proposed Experimental Design

The first broad binary experiment should answer three questions:

1. Does the principled local logistic mode improve probability prediction over the minimal Brier/least-squares mode?
2. When logistic mode fails or falls back locally, can telemetry explain the affected cases?
3. Are the practical conclusions stable across dimension, geometry, event-rate profile, sample size, and signal strength?

8.1 Methods To Compare

The initial method roster should include:

- **LPS-Brier:** `fit.lps(outcome.family = "bernoulli").`
- **LPS-logistic:** `fit.lps(outcome.family = "binomial").`
- **Oracle truth reference:** not a fitting method, but the known probability surface $p(x)$ used to compute synthetic evaluation metrics.
- **Optional classical baselines:** logistic generalized linear model, random forest probability model, and gradient boosting probability model, if runtime permits.

The first comparison should keep ordinary LPS and logistic LPS as the main scientific contrast. Additional baselines are useful, but they should not obscure the main implementation question.

8.2 Dataset Families

The experiment should use synthetic binary outcomes over frozen geometries already used in recent P7X-style LPS experiments:

- one-dimensional and low-dimensional embedded coordinate examples;
- non-manifold examples such as intersecting components or mixed local dimension;
- VALENCIA-derived relative-abundance or dCST-derived geometries;
- high-dimensional observed-coordinate examples where local chart dimension estimation matters.

For each geometry, define a probability surface $p(x)$. Then generate

$$Y_i \sim \text{Bernoulli}\{p(x_i)\}.$$

8.3 Probability-Surface Factors

The probability surfaces should vary along interpretable axes:

- **Event prevalence:** low, moderate, and high average event rate.
- **Boundary sharpness:** smooth logistic ridge versus steeper transition.
- **Spatial complexity:** one bump, two bumps, and ridge-mixture probability surfaces.
- **Class imbalance:** cases where event rates are near 5%, 20%, and 50%.

The design should avoid unrealistic spike-like profiles as the default, especially for microbiome-style applications. Mixtures of smooth Gaussian-like components and smooth logistic transitions are better first targets.

8.4 Tuning And Shared Candidate Policy

To compare the two binary modes fairly, both should use the same candidate families where possible:

$$k \in \mathcal{K}, \quad q \in \mathcal{Q}, \quad K \in \mathcal{K}_{\text{kernel}},$$

where k is support size, q is polynomial degree, and K is the kernel. The first broad run should use the same default design basis and chart policy as the current continuous-response LPS winner unless the binary smoke telemetry shows that logistic fitting is unstable.

For scoring:

- LPS-Brier should select candidates by CV Brier score.
- LPS-logistic should select candidates by CV log loss.
- Both methods should report both Brier score and log loss so the comparison is not dependent on one score alone.

8.5 Evaluation Metrics

Because the probability surface $p(x)$ is known in synthetic experiments, the primary metric should be probability RMSE:

$$\text{TruthRMSE}_p = \left[\frac{1}{n} \sum_{i=1}^n \{\hat{p}(x_i) - p(x_i)\}^2 \right]^{1/2}.$$

Additional metrics should include:

$$\begin{aligned} \text{Brier}_{\text{obs}} &= \frac{1}{n} \sum_{i=1}^n \{Y_i - \hat{p}(x_i)\}^2, \\ \text{LogLoss}_{\text{obs}} &= -\frac{1}{n} \sum_{i=1}^n [Y_i \log \tilde{p}_i + (1 - Y_i) \log(1 - \tilde{p}_i)], \end{aligned}$$

where $\tilde{p}_i$ is the numerically clipped probability used for log loss. Calibration diagnostics should be added after the first broad run if the two modes show similar TruthRMSE but different probability sharpness.

8.6 Failure And Fallback Accounting

Every report should include unavoidable accounting rows:

- planned tasks;
- successful tasks;
- non-finite tasks;
- error tasks;
- timeout tasks;
- logistic fallback fraction, separated into CV and final fitting.

For logistic LPS, the final report should stratify performance by fallback fraction. A method that wins only because many local logistic fits fell back to event-rate averages should be interpreted differently from a method whose local logistic solves converged cleanly.

8.7 Recommended First Run

The recommended first broad run is a smoke-scale factorial design:

- 6 to 10 frozen geometries;
- 2 event-rate levels;
- 2 smoothness/transition-sharpness levels;
- 3 random response replicates per case;
- both LPS-Brier and LPS-logistic;
- hard per-task timeout and failure isolation.

This is large enough to reveal obvious instability, but small enough to revise before launching a full overnight benchmark.

9 Planned Results Addendum

The next revision of this progress report should add:

- a compact table of planned versus successful tasks;
- a paired comparison of LPS-logistic minus LPS-Brier TruthRMSE;
- a paired comparison of observed log loss and observed Brier score;
- a figure showing logistic fallback fraction versus TruthRMSE regret;
- a short decision paragraph saying whether logistic mode is ready for broader PS-LPS binary development.

10 Current Recommendation

The current recommendation is to keep both binary modes. The "**bernoulli**" mode is the safer baseline because it is close to ordinary LPS and estimates the conditional expectation directly under Brier risk. The "**binomial**" mode is the more principled probability model and should be tested seriously, but its broader use should depend on fallback telemetry and runtime behavior.

The immediate next step is not to modify PS-LPS. It is to run the proposed LPS binary experiment first, decide whether local logistic LPS is stable and useful, and only then design a synchronized binary PS-LPS objective.